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RESEARCH-ARTICLE

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Application of big data in construction of a book management model

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Abstract

Big data mining techniques are the primary technological means for extracting the inherent patterns from massive information, and they have currently gained widespread application. This paper analyzes the system architecture of existing library management systems and explores how to add a new data mining module on the basis of existing library management systems, with the primary focus being the interface issue between the new data mining module and the original system. A data mining module structure based on Web Services is designed, where the invocation of the new data mining sub-module is realized through residing a stub in the original system. The original data of the library management system is analyzed, and the data is re-extracted, integrated, and standardized to meet the requirements of data mining. A data model compatible with mining is also designed to facilitate the improvement of mining efficiency. An analysis was conducted on personalized services and management decisions for readers, and a book recommendation service based on readers' reading interests was proposed to improve service quality. Through association rule mining and cluster analysis, intelligent decision-making in the book management process was achieved, improving management quality.

CCS Concepts

• **Network** → Network algorithms; Data path algorithms; Packet classification.

Keywords

Big data, Data mining, Library management, Association rules, clustering algorithm

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1 Introduction

With the development of information technology, various information resources have become an indispensable part of social production and life, and even one of the most critical resources [1]. In recent years, the development of the Internet and information industry, as well as the ubiquitous information perception and collection terminals, have enabled us to collect massive data, leading to an exponential growth in the volume of information materials across many industries. McKinsey, a renowned global consulting firm, stated: "Data has permeated every industry and business functional area today, becoming a crucial factor of production. The mining and utilization of massive data herald the advent of a new wave of productivity growth and consumer surplus." Faced with this massive data, it is of paramount importance to unearth the valuable information or knowledge hidden behind these massive data [2].

MyLibrary is a relatively typical personalized service model, widely applied in library management systems [3]. Its primary objective is to assist readers in selecting and reorganizing information resources of interest from the resources provided by the library management system, storing them in MyLibrary. Subsequently, readers can access their preferred information resources through MyLibrary, effectively reducing the amount of information retrieval and enabling personalized information recommendations and customized information services [4-6]. Currently, many universities utilize this system to achieve personalized services. For instance, Cornell University Library's MyLibrary@Cornell system, the MyLibrary@LANL system of North Carolina State University, etc. Among these, the MyLibrary@Cornell system primarily offers three service functions: MyLinks, MyUpdates, and MyContents. MyLinks is a service for readers to organize and manage digital resources; MyUpdates is a system that periodically discovers new resources and notifies readers, enabling them to incorporate these resources into their library management system; MyContents is a service that recommends journal contents to readers [7].

The construction of personalized service system of digital libraries in China started late, but has also made good achievements. Libraries in various universities are building My library system to realize personalized information services of library information resources. For example, Peking University Library, Tsinghua University Library, Renmin University of China Library, Zhejiang University Library and Huazhong University of Science and Technology Library are all building personalized library service systems, which have achieved good results.

In China, the construction of personalized service systems for digital libraries started relatively late, but good progress has been

made. Libraries in various universities have been building My Library systems to realize personalized information services for library information resources [8]. For example, Peking University Library, Tsinghua University Library, Renmin University of China Library, Zhejiang University Library, and Huazhong University of Science and Technology Library have all been building personalized library service systems, and have achieved positive results [9]. University libraries are mainly intended to improve the level of university education and the competitiveness of universities. They aim to build libraries of books, documents, and other resources that embody professional characteristics, and continuously improve and supplement them based on the development of the university and the actual situation of students. They also conduct effective resource elimination and replacement to make them more conducive to discipline construction. In addition to these explicit resources, there are also hidden information resources, namely, a large amount of circulation information accumulated during the long-term operation of the library management system. For example, this information includes Web logs, circulation logs, and access records of various resources in the automated university library management system [10]. The accumulation and utilization of this information is more conducive to various decision-making in library management. Therefore, this accumulated information lays a solid foundation for subsequent data mining work. Most universities already have their digital libraries and have achieved automation.

This research is based on a data mining model for the library management system. A data mining web service is established to perform data mining on relevant information within the library management system, implementing two main functions: web information service and data mining. These functions reside on separate servers to reduce the load on the web server. Web information service and data mining operate independently through a loosely coupled manner, ensuring that issues with the data mining module do not affect the normal operation of other library system functions. Clustering algorithms are employed to perform reader clustering and book clustering respectively. Reader clustering primarily involves analyzing the main literature needs of different reader categories, laying the foundation for future book recommendations and providing decision support for book acquisitions. Book clustering focuses on analyzing book lending volumes to identify different book categories.

2 Data mining algorithm

Data clustering, as a major tool in data mining, can obtain the distribution of data through clustering analysis, such as identifying dense and sparse regions, discovering global distribution patterns, and understanding the inherent interrelationships between data. The commonly used data structure in cluster analysis is the dissimilarity matrix, also known as the object-object structure, which represents the similarity between n × n stored objects pairwise and is represented as an n dissimilarity matrix. Its structure is as follows:

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ d(2,1) & 0 & 0 & 0 \\ d(3,1) & d(3,2) & 0 & 0 \\ d(n,1) & d(n,2) & 0 & 0 \end{bmatrix} \tag{1}$$

In which d(i, j) represents the dissimilarity between objects i and j. It is typically a non-negative number, where a smaller value closer to 0 indicates higher similarity between the two objects i and j, while a larger value signifies greater dissimilarity. This type of dissimilarity matrix is known as a one-mode matrix because its rows and columns represent the same entities. Therefore, when i = j, d(i, j) = 0.

Before measuring the dissimilarity of objects, it is necessary to standardize the dissimilarity data. After data standardization, the distance measurement between objects is calculated, and their dissimilarity is determined based on their distance. The method used for data standardization is to convert the original measurement values into unitless values. The distance measurement between objects mainly uses Euclidean distance, Manhatan distance, and Minkowski distance.

The Euclidean distance is defined as formula (2):

$$d(i,j) = \sqrt{|x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + ... + |x_n - x_{jn}|} \tag{2}$$

Where $i = (x_{i1}, x_{j1}, ..., x_{in}), j = (x_{j1}, x_{j2}, ..., x_{jn})$is two n-dimensional objects to be calculated.

The k-means algorithm takes k as a parameter to partition n objects into k clusters, such that objects within each cluster have high similarity while the similarity between clusters is low.

The Apriori algorithm is a single-level, single-dimensional, boolean association rule mining technique. It is one of the most influential algorithms for mining frequent boolean association rules. It adopts a level-wise, iterative search approach. First, it finds the frequent 1-itemsets (L1). Then, based on L1, it computes the frequent 2-itemsets (L2). From L2, it derives the frequent 3-itemsets (L3), and so on, until no more frequent k-itemsets (Lk) can be found. Finding each Lk requires one database scan. The key aspects are that the algorithm is single-level (does not consider multi-level item/attribute abstraction), single-dimensional (processes one attribute/dimension at a time), and boolean (handles boolean association rules on discrete item values).

The Apriori algorithm is accomplished by followings:

Input: Transaction database T, minimum support minsup.

Output: Frequent itemsets L in T.

$$\begin{aligned} &L_1 = (\text{Frequent1} - \text{itemsetsinT}); for(k = 2; \\ &L_{k-1} \neq \phi; k++)\{C_k = appriori_gen(L_{k-1}, \\ &\min\sup); foreachtransactiont \in T\{C_t = \\ &subset(C_k, t); foreachcandidatec \in C_t \\ &c.count++; \}L_k = \{c \in C_k | c.count \geq \min\sup\}; \\ &\}returnL = \cup_k L_k \end{aligned}$$

Data cleaning. Data cleaning mainly attempts to fill in missing values, identify outliers, eliminate noise, and correct data inconsistencies. The data is mainly smoothed using methods such as box sorting, clustering, computer and manual inspection combined with regression. Data integration. In the process of implementing data integration, the main consideration is data pattern matching and data redundancy.

Data transformation. Smoothing: It is the process of removing noise from data to avoid significant interference from a small amount of noisy data on mining results. The main methods used are smoothing and clustering generalization. Data reduction. In the process of data mining, if the dataset is too large, it will affect

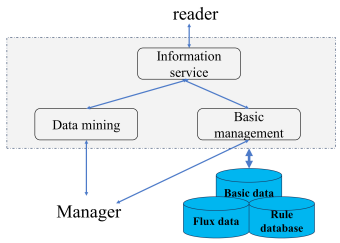

Figure 1: The functional structure of a book management system based on data mining.

the speed of data mining. Data reduction is to obtain the compression result of the dataset, which is much smaller than the original dataset, but still close to maintaining the integrity of the original data. This way, the results of data mining are more effective and can produce the same or almost the same analysis results.

The data mining-based library management system proposed in this paper mainly establishes a data mining functional module and a data mining service on the basis of the original library management system. By invoking these web services, more personalized services can be provided to readers. At present, there are three main data mining services established. (1) Reader data clustering service (ReaderKmeans). This service performs cluster analysis on readers based on their borrowing volume, classifies all readers, and identifies different reader groups, which facilitates the overall analysis of readers. (2) Book clustering management service (BookKmeans). This service categorizes books according to their borrowing volume, identifies book categories that are more popular with readers and book categories that are rarely borrowed, and provides support for future acquisitions. (3) Association rule mining service for borrowing information (LibApriori): This service mines the association relationships between borrowed books, the relationships between borrowed books and professional categories, and the relationships between borrowed book categories and professional categories based on readers' borrowing records. It provides decision-making information for book recommendations, book purchases, and book shelving.

3 Experimental Analysis

The library management system proposed in this article provides users with a functional structure as shown in Figure 1. Readers can access the information services on the WEB site to obtain the books and materials they need. At the same time, this information service will automatically call the data mining module based on the reader's relevant information to recommend books and materials that readers are interested in. For administrators, they can not only call the basic management interface to implement basic circulation management of books, but also implement some data mining function modules to mine management decision information.

Incorporating a data mining module into a library management system primarily aims to support scientific decision-making during the book management process. This deviates from the traditional library management systems that only perform simple data statistics. By enabling in-depth information mining within the library management system and leveraging data mining techniques to analyze reader demands, readers can be categorized into different groups based on their professions. This allows for predicting reader borrowing behaviors. Ultimately, personalized recommendations can be provided according to the diverse needs and personal interests of different readers. Additionally, it can mitigate decision-making errors and other management issues in the book management process. The key functions are as follows:

Mining the inherent borrowing connections of different categories of readers mainly refers to which categories of books different types of readers tend to borrow during the book borrowing process. Since the books that these readers are interested in may be in the same professional category or different professional categories, this association rule mining can identify the categories of books that readers are interested in. This can provide personalized services for different categories of readers. Mining the association rules of borrowing by readers of different majors can uncover the inherent correlation between different disciplines in the transaction database. For example, some readers borrow books in major A and books in major B at the same time. By conducting association analysis on the readers' borrowing information and various disciplines, the implicit reader borrowing interests can be identified. The layout of the library collection can be designed and adjusted, and books with strong correlations can be placed in close proximity to make it easier for readers to find professional books that match their interests. This is different from the current distribution status, where books on computer science need to be found in the natural science library, while books on foreign languages need to be found in another library, which improves the utilization rate of books and periodicals.

Book clustering analysis. This mainly involves mining the inherent relationships between borrowed books based on their basic borrowing information. Through data mining, the library management system can identify the books that readers prefer to borrow as well as those that are rarely borrowed. This analysis provides support for subsequent book acquisitions.

Reader clustering analysis. The main approach is to reclassify readers based on their borrowing information, not just based on their department or major. For example, computer science students may prefer to borrow books related to literary works, and therefore reclassify readers to provide support for book procurement and personalized services for readers.

The data mining based book management system proposed in this article mainly implements data mining functions on the basis of the original book management system, and adds new functions without changing the original book management functions. This can enable various universities to add a module to the existing library management system without the need to change it, which not only saves costs but also enables personalized services. The basic structure of the system is shown in Figure 2.

The user layer mainly refers to the various types of users participating in the system, primarily readers and administrators. In this context, reader users are categorized into professional teachers, students, and other researchers. Students are further classified

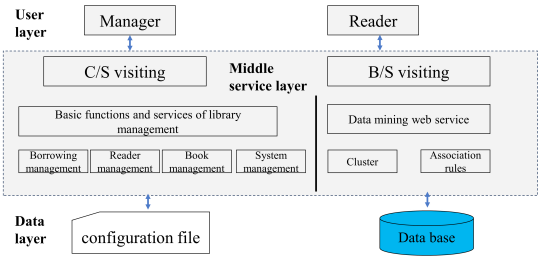

Figure 2: Architecture of library management system based on data mining.

based on their academic disciplines to facilitate subsequent user clustering.

The middle service layer is the primary module involved in logical processing. To enable convenient system access for different users, the system employs two access points: a Client/Server (C/S) access point and a Browser/Server (B/S) access point. The C/S access point is designed for system administrators, librarians, and other staff members, who can manage the system, manage books, handle borrowing, manage readers, and perform other basic administrative tasks by installing the library management system client. The B/S access point is designed for readers, allowing them to access the information server through a web browser and perform operations such as book searches, borrowing status inquiries, and using personalized "My Library" features.

Both access points can invoke the data mining Web Service, calling clustering and association rule mining algorithms to perform reader clustering, book clustering, and mining of borrowing information association rules. This achieves the provision of personalized services to users.

The data layer primarily consists of various data in the library management system, including basic data such as book data, reader data, borrowing data, and system configuration files.

Data mining is the process of analyzing and processing relevant datasets to identify their internal patterns, which requires a significant amount of system resources. Therefore, if data mining services and basic library management services are deployed on the same server, it will inevitably increase the load on the server. As a result, with a large number of users accessing simultaneously, the system's response speed will be slow. Therefore, in improving the personalized service level of the library management system, efforts should also be made to minimize the impact on the system's response speed.

After analyzing the aforementioned issues, this paper proposes a data mining module based on Web Services, constructing Web Services to implement clustering and association rule mining services. The Web service protocol stack, from bottom to top, consists of six layers: network layer, transport layer, message exchange layer, service description layer, service discovery layer, and application layer. The message exchange between the server and client occurs at the application layer, and this message exchange requires the support of the underlying protocols. The service discovery layer protocol is primarily used for registering, finding, and locating various Web services. The service description layer mainly implements detailed descriptions of services and provides them to directories. The message exchange layer is the mechanism for inter-process communication, including message encapsulation. The transport layer is responsible for message delivery, while the network layer refers to the actual transmission and routing of data packets.

Web services are network services delivered via HTTP that can leverage existing distributed components to form network applications. These distributed components are language- and platform-independent, loosely coupled, and programmable based web applications that can be described, published, discovered, coordinated, and configured using open XML standards. According to the W3C definition, a web service is a software system designed to support interoperability between different machines on a network. Network services are typically composed of multiple application programming interfaces that execute service requests submitted by clients through the remote server side of the network.

SOAP (Simple Object Access Protocol): SOAP is a lightweight protocol for exchanging XML-encoded information. It is an XML-based extensible messaging format that requires binding to a transport protocol such as HTTP or SMTP. It has three main aspects: the XML envelope, which defines the framework for describing the content of the message and how it should be processed; the rules for encoding procedure calls into XML objects; and the conventions for executing remote procedure calls (RPCs).

WSDL (Web Service Description Language): WSDL is an XML-based language for describing web services, their functions, parameters, and return values. It is used to describe the details of how to access a service endpoint and the protocol to use. This allows service clients to set the appropriate parameters and access the service based on its description document when requesting access to the service. Since WSDL uses XML for description, it is easy for machines to parse.

UDDI (Universal Description, Discovery, and Integration): UDDI is a protocol for publishing and searching for web services. Applications can use this protocol to find target web services at design time or runtime. It is a set of web-based, distributed, and standardized specifications for providing an information registry for web services. It also includes a set of standardized specifications for access protocols that allow businesses to register the web services they provide so that other businesses can discover them.

In the Web Service work mode, the service provider needs to register itself with the directory server. This way, when a process that needs a specific service is running, it can contact the directory server to search for the specified service that the process needs. If the service is available, the directory server will return the relevant description information for the service (mainly some detailed invocation parameters). With this description information, the process can interact with the service.

Systems that adopt a Web Service-based architecture are loosely coupled, which makes it easy to add or remove functional modules from the library management system. In addition, since the daily management of books and the data mining function are separated into independent servers, the execution of the data mining function does not affect the normal operation of the original library management function. At the same time, using different servers will not

Table 1: Mining Association Rules for Book Borrowing Information.

Number	Association rules	Confidence level
1	978-7-81081-964-0=1==>978-7-107-18724-7=1	confidence=2/2=1
2	978-7-107-18724-7=1==>978-7-81081-964-0=1	confidence=2/3=0.67

increase the load on the web server, ensuring that the system does not deteriorate in performance or even crash after adding heavy data mining functions.

The Apriori algorithm is used to mine the correlation between book borrowing information, mainly based on the rules between reader's book borrowing information to achieve book recommendation services. The book recommendation methods proposed in this article mainly have two types: system active recommendation and reader browsing recommendation.

The association rule mining of reader borrowing information (the intrinsic relationship between books) mainly aims to explore the intrinsic relationship between borrowed books by selecting a reader borrowing dataset. This article selects the top 1000 books in the book borrowing ranking of a certain university in 2023 for mining. For example, Table 1 shows some borrowed data, randomly selecting 20 borrowing records from 10 books. For the storage space of the system, it is relatively wasteful, and the retrieval space of the algorithm is larger, which consumes more time. Moreover, with the increase of books, reaching several thousand copies, each student may only borrow a few books at a time, which obviously leads to significant consumption of storage resources and algorithm scanning time. Therefore, this article adopts a new sparse data processing method to reduce the consumption of system storage and computing resources Table 1.

Based on the analysis of the results in the table, obtain the basic borrowing patterns of readers. Readers who borrow books with ISBN numbers 978-7-81081-964-0 will also borrow books with ISBN numbers 978-7-107-18724-7. The probability of readers borrowing books with ISBN numbers 978-7-107-18724-7 is 67%. Therefore, the system recommends in the early stage. The data mining system will scan the reader's borrowing record library. According to association rule 1, if a reader's borrowing record contains books 978-7-81081-964-0 but no books 978-7-107-18724-7 appear, they will be recommended to the corresponding reader. Reader browsing recommendations. When readers are searching for books, based on the books entered by the readers, according to association rule 1, if the readers input books 978-7-81081-964-0, the corresponding books 978-7-107-18724-7 are recommended for readers to browse, reducing their search time.

Using the k-means clustering algorithm to cluster and mine the borrowing volume of books, in this algorithm, the borrowing volume of books should be divided into three types: popular books with high borrowing volume, general books, and books with relatively low borrowing volume, mainly to provide support for future procurement. For hot books, it is necessary to increase the purchasing volume, while for books that are not sought after, the purchasing volume should be reduced or not made. The clustering results in Table 2 indicate that the books in Cluster C1 have relatively small borrowing volumes, with an average borrowing frequency of 16 times; A book with a relatively large borrowing volume has an average borrowing frequency of 193.1667 times. Therefore, in the book procurement plan, the purchase volume of books in the C3 cluster should be increased based on this clustering algorithm. And reduce the purchase volume of books in C1 cluster.

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4 Conclusion

Data mining technology is the main technical means of acquiring the hidden规律 within massive information, and it has been widely used at present. Currently, with the changes in the scale and goals of higher education, library information resources are also continuously growing. The hidden information contained in a large amount of circulation information is the key information to realize the personalized service of the library management system. Therefore, mining the circulation information within the library management system is conducive to improving the service quality and management quality of the library management system and improving the utilization rate of books.

In this study, multiple servers are established to separate intelligent services such as data mining from the basic services of the library management system. Web Services technology is adopted to enable information flow between different service hosts, mitigating the issue of reduced reader information retrieval performance caused by data mining computations. Through an analysis of the existing library management system's data, Web services are designed to implement data mining services, enabling personalized services and scientific management decision-making services. The association rule mining algorithm is employed to uncover the association relationships between book borrowing information, identify borrowers' interests in book borrowing, and provide targeted book recommendations. Simultaneously, reader clustering based on borrowing volume and book clustering based on borrowing volume are performed, and association rule mining is conducted for book categories and borrowing patterns of readers from different academic disciplines. This approach provides information support for formulating book acquisition plans, book shelving, and other management tasks.

Table 2: Cluster mining results of book borrowing volume.

Cluster center	Clustering
C1=0.00433	C1 ={978-7-111-25662-5,978-7-04-021789-6,978-7-80000-804-7,978-7-80206-509-3, 978-7-229-00284-8,978-7-300-09202-7,978-7-312-02246-3,978-7-03-021112-5}
C2=0.269	C2={978-7-122-04194-4,978-7-307-06669-4,978-7-04-024711-4,978-7-81107-727-8, 978-7-5609-4685-6,978-7-214-05366-4}
C3=0.7408	C3={978-7-5635-1377-2,978-7-81117-507-3,978-7-222-05568-1,978-7-229-00044-8, 978-7-5331-4689-4,978-7-5352-3945-7}

At present, association rules are used to implement book recommendations, but for university library management systems, there are also many important digital resources, such as various digital resource libraries. These resource libraries are queried according to the originally defined query mode, and no analysis and recommendations are made based on the reader's query history. Therefore, implementing the recommendation service for such resources will be an urgent research direction in the future.

This paper uses Web Services technology to achieve loose coupling between the new data mining module and the original system module, and realizes the plug-and-play addition and exit of the system module. However, this consumes system resources to a certain extent. Therefore, with the development of cloud computing and big data processing technologies, researching the use of new technologies to solve the problem of loose coupling between systems will be a major goal in the future.

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